Sharjeel Ahmad

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Education

University of Illinois Urbana-Champaign (UIUC) **Aug 2024 – Present**
*Master of Science in **Physics*** *Urbana-Champaign, Illinois*

cGPA: **3.95**/4.00

Graduate Courses: Condensed Matter Physics, Modern Computational Physics, Quantum Information and Optics, Quantum Information Theory and Processing, Quantum Physics, Representation Theoretic Methods in QIT

Lahore University of Management Sciences (LUMS) **Sep 2020 – Jun 2024**
*Bachelor of Science **Physics Major, Mathematics Minor*** *Lahore, Pakistan*

Research Experience

Graduate Research Assistant **May 2025 – Dec 2025**
Supervisor: Dr Eric Chitambar *UIUC*

- Led a summer research team developing the **Embedded Dense Coding** protocol, solving numerical optimization problems (MATLAB CVX) to show one-way teleportation fidelities surpassing classical limits via the **teleportation/dense-coding duality**
- Helped design Qiskit simulations on **IBM Torino** to evaluate EDC performance under native gate noise, demonstrating enhanced success probability compared to standard dense coding
- Co-authored the EDC technical write-up, presented a **poster** at the **IQUIST All-Hands Poster** Session, and gave a **research talk** at the **Qiskit Fall Festival**, to showcase our research findings
- Currently finalising a manuscript to be submitted for peer review

Graduate Research Assistant **Aug 2024 – May 2025**
Supervisor: Dr Paul Kwiat *UIUC*

- Developed single and **multi-depth computer-generated holograms** using MATLAB and the Holoeye SLM Pattern Generator, optimizing Fourier-transform algorithms for high-resolution film reconstructions
- Built and aligned fully **portable optical setups** and operated the Lasergraphics Mark III film recorder to print, test, and troubleshoot holograms
- Co-authored a comprehensive **instruction manual**, establishing a complete documentation pipeline that preserves experimental methods and supports future students in the Kwiat group

Graduate Research Assistant **Aug 2024 – May 2025**
Supervisor: Dr Kejie Fang *UIUC*

- Submitted a **report** to experimentally demonstrate **quantum teleportation** via a **nanophotonic** nonlinear Bell state analyzer on a **4-dimensional qubit**
- Conducted experiments for telecom **photon wavelength conversions** under low power conditions using **Indium gallium phosphide** (InGaP) micro-waveguides
- Conducted fidelity derivations on **Heralded Entanglement Photon Generation** on SPDC systems and Non-linear Sum-Frequency Generation systems

Senior Year Project **Spring 2024**
Supervisor: Dr Adam Zaman *LUMS*

- Did a **literature review** on the **Lindblad Master Equations** and **Liouville's Theorem**
- Applied a variation of Kubo Formulation to study the **Time Evolution** of particular **Open Quantum Systems** such as driven **Spin-Boson** systems
- Studied various **Perturbation Theories** which are commonly applied in Open Systems as well as **Heat Transport** equations

Research Assistant **May 2023 – May 2024**
Supervisor: Dr Ammar Ahmed *LUMS*

- Collaborated with the life sciences department to utilize the **Nematic Liquid Crystal '5CB'** to differentiate PPE and PPP plastics found in the local environment through **Optical Birefringence**

- Submitted a presentation which analyzed the collected data through **Rotational Analysis** in a **Polarized Microscope** on the plastic samples coated with 5CB
- Created a roughness profile of the various plastic samples by taking **2D Fast Fourier Transforms** of the captured images in **MATLAB**

Topological Defects and Photo-lithography in Liquid Crystals — **Spring 2021 – Spring 2023**
Research Assistant under Dr Ammar Ahmed — *LUMS*

- Conducted experimental research on **Lyotropic** and **Non-Lyotropic Liquid Crystals** and their applications in bacterial transport
- Employed **Auto-CAD** to generate micropatterns on a photo-lithography mask for the **Functional Materials and Optoelectronic Devices** group which is still used today

Professional & Volunteer Work

Pakistan National Debate Team — **Jan 2023 - Jul 2024**
Coach — *Lahore, Pakistan*

- Selected to to train and **manage** the National Team to **Serbia WSDC 2024** held in July
- Streamlined the team selection process by creating a Python algorithm to collect and analyze **36 unique data points** across **40 shortlisted applicants** to sort them by their weighted Z-scores
- Held regular training camps to improve the team's persuasive speaking skills by focusing on strategic thinking, enabling them to reach the Octafinals of WSDC 2024

Community Service Society - Bloodlink — **Fall 2021 – Fall 2022**
Volunteer — *Lahore, Pakistan*

- Introduced a **Digitization Framework** to reduce response time and alleviate the need for manual oversight
- Maintained a **24-hour helpline** on designated days, connecting hundreds of patients with blood donors

Honors & Awards

Dean's Honor List — **Fall 2020 – Fall 2024**

- Awarded for maintaining a **cGPA greater than 3.6** in each semester — *LUMS*

Chandrasekhar Honorific Fellowship — **Fall 2022 – Fall 2024**

- Awarded twice for **academic excellence** in physics from a cohort of 60 candidates — *LUMS*

50% Merit Scholarship — **Fall 2020 – Fall 2022**

- Awarded for being in the **Top 10 freshman and sophomore undergraduate students** in the LUMS School of Science and Engineering

World Schools Debating Championship — **Jul 2019, 2020**
Two time Member of national team — *Thailand, Online*

- Selected to serve in the **National Team** out of a pool of **300 candidates** from all across Pakistan
- Ranked as the **8th Best Speaker in the world** in the English as Second Language Category

Course Projects & Presentations

Post-selection techniques and applications in QKD and Channel Coding — **Fall 2025**
MATH 595 - Representation Theoretic Methods in Quantum Information Theory

Photonic Quantum Computing — **Spring 2025**
PHY 513 - Quantum Optics and Information

Resource Theory of non-Gaussian Operations — **Fall 2024**
ECE 498 - Quantum Information Processing & Theory

On Recurrent Neural Networks in Unsupervised Deep Learning — **Fall 2023**
PHY 603 - Machine Learning for Physics

On Path Integral Quantization in Scalar, EM and Spinor Fields — **Spring 2023**
PHY 539 - Introduction to Quantum Field Theory

Skills & Interests

Skills: Python (Qiskit), MATLAB (CVX), LaTeX, Microsoft Office, Auto CAD
Interests: Public Speaking, Tennis, Futures Trading, Fantasy Novels